

MATSim User Meeting 2026

A Crowdsourced-First Calibration Framework for Multi-Class MATSim Models in Data-Constrained Heterogeneous Traffic

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Outline

1

The problem

Mixed traffic. Sparse counts. Abundant travel times.

2

MATSim implementation

Scenario, QSim, CaDyTS, event listeners

3

The calibration loop

Score, adjust, roll back

4

Three objectives

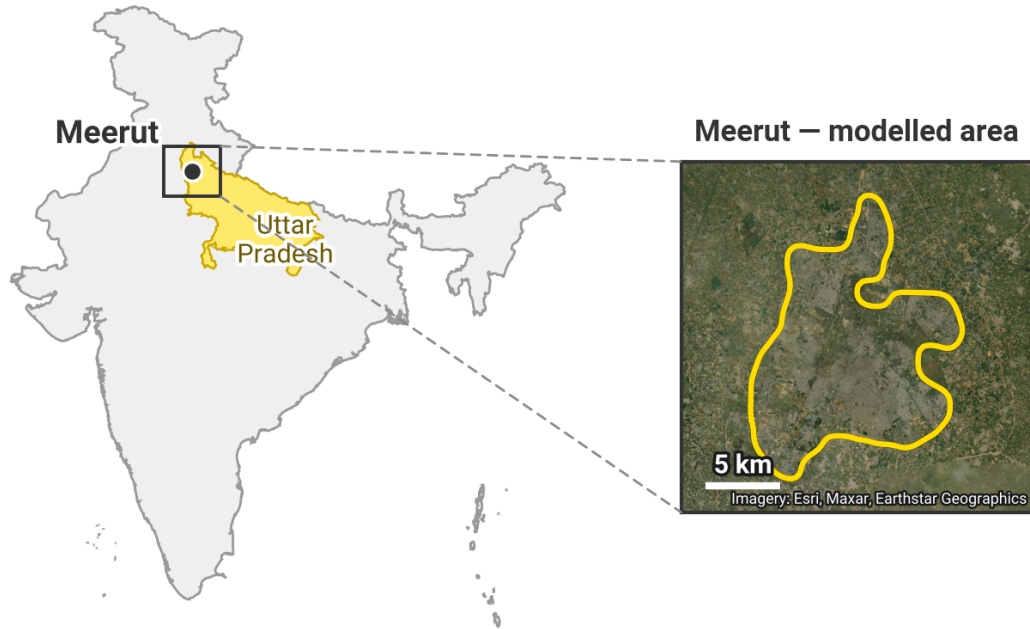
Volume-based, travel-time-based, multimodal

5

Results and what they imply

Which objective wins where, and what it costs to collect

Study area



TRAFFIC ANALYSIS ZONES

74

OpenStreetMap network

MOTORCYCLE SHARE

53.2%

Car 19.4% · auto 18.5%

MOTORISED CLASSES

7

Motorcycle to truck

COUNTED LINKS

100

At 28 intersections

Classified counts

100 links at 28 intersections, by vehicle type

Absolute flow by vehicle class. Collected for each approach.

Crowdsourced travel times

100 of the same links, every hour · R^2 0.69–0.83

Hourly through the day. Weak where queues form.

Network

Built from OpenStreetMap

26,524 nodes · 69,547 links

74 zones, seven fixed-mode populations

EPSG:32643, refined for local topology

Both datasets sit on the same approaches

100 counted links at 28 intersections

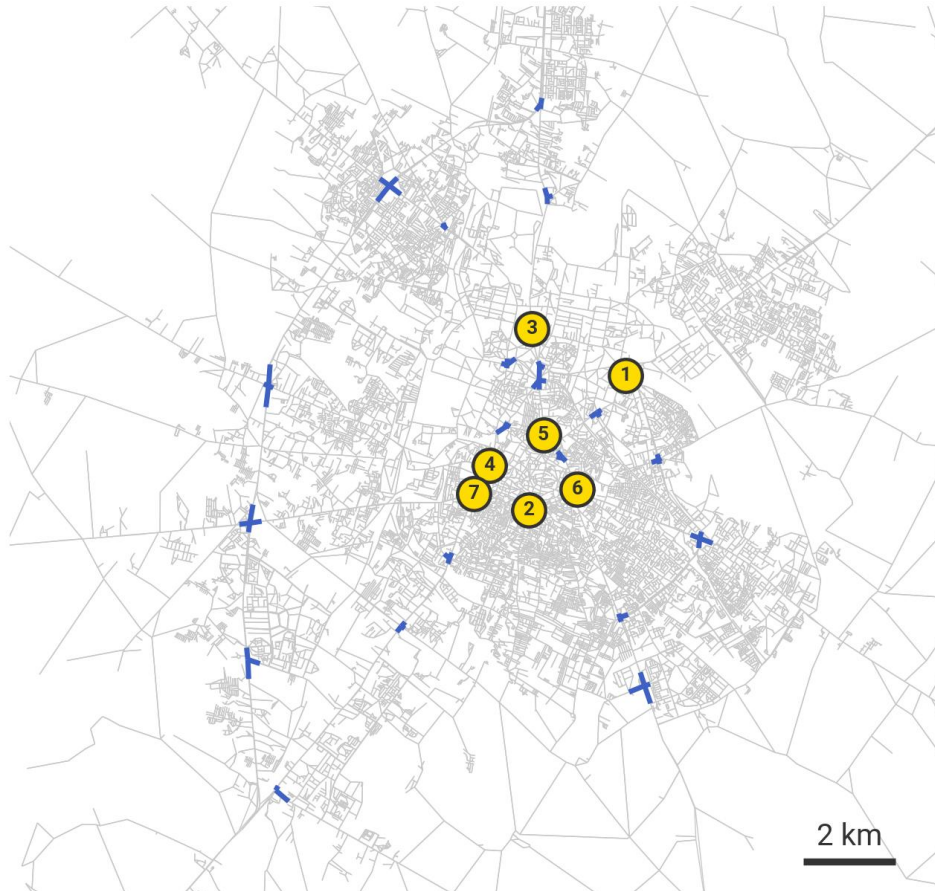
100 of them with hourly travel times and volumes

Seven intersections carry the congestion

Ranked by morning-peak approach delay

Calibrated from volumes

Everything else stays on travel time



- Modelled network
- Calibration approaches
- Seven most congested intersections

Implementation

Scenario	Plan selection	Event listeners
• QSim, queue traffic dynamics • Seven fixed-mode populations • Volume-based OD: video counts, MLE • Travel-time OD: Rapidex • Motorcycles on seepage dynamics	• Charypar-Nagel scoring • Stuck-agent penalty • CaDyTS link-count correction • SelectExpBeta plan choice • ReRoute early, then disabled	• Link volumes, aggregate and per mode • Travel times and delays by hour • Stuck agents and zero-flow links • Network state written each round

Calibration loop

1

Simulate

MATSim iterations on the current OD and network

2

Score

Listener output becomes travel-time, volume and delay error

3

Select

The best state in the round is retained

4

Adjust

Mode ODs regenerated. And minor capacity adjustments.

5

Damp

Change is bounded. Deterioration rolls back and halves the step.

The loop never loses its best state.

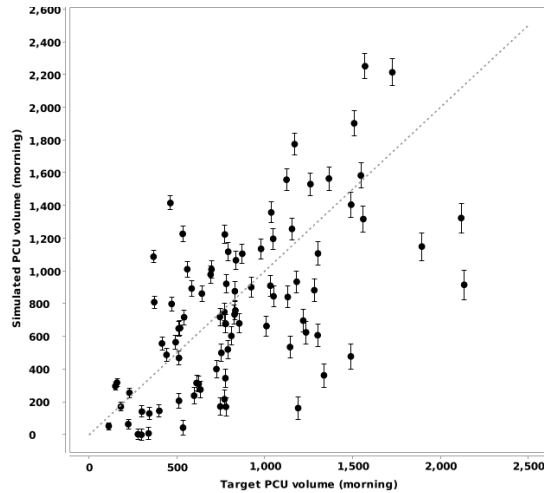
A bad adjustment costs one round, not a campaign.
Every round is reproducible from its retained state.

Objectives

	Volume-based	Travel-time-based	Multimodal
Demand source	Video counts, MLE	Rapidex	Inherited
Evidence	Counts	Travel times	Counts at seven intersections, travel times elsewhere
Target	85% at GEH < 10	85% within 20% or 10 s	Composite of both
Network levers	Capacity, lanes, speed	Capacity, lanes, speed	Frozen
Cross-metric	Travel time as diagnostic	GEH penalty in score	Both inside the score

Baseline results

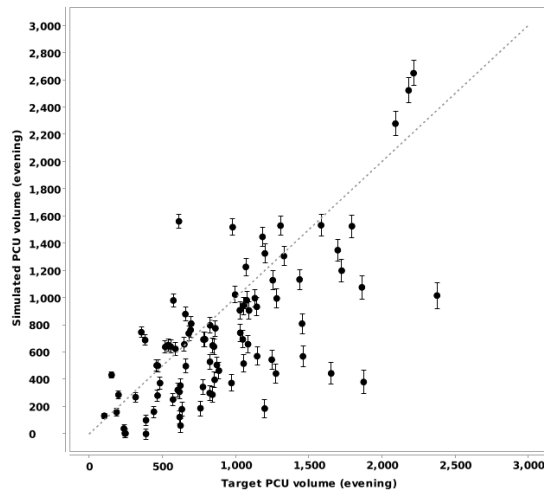
Volume-based simulated vs target PCU volume



Morning peak

10:00 – 11:00

GEH < 10 on 46.6%
of 88 observations

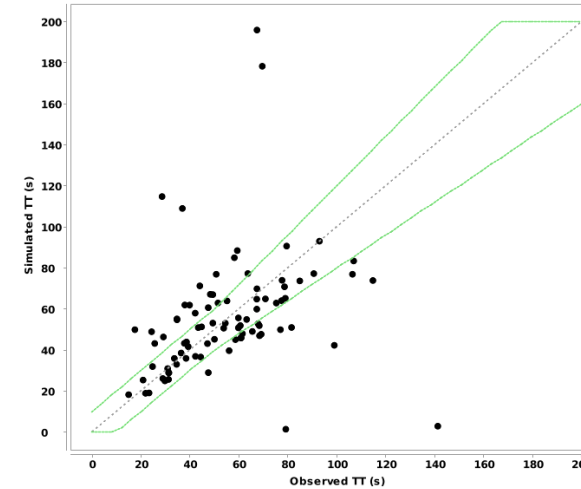


Evening peak

16:00 – 17:00

GEH < 10 on 52.3%
median GEH 9.3

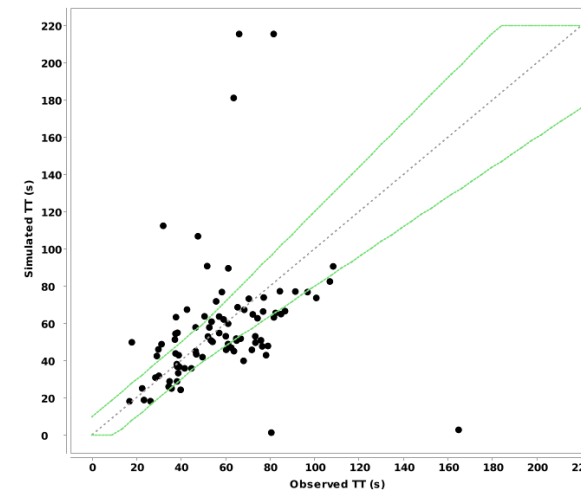
Travel-time-based simulated vs observed travel time



Morning peak

10:00 – 11:00

Criterion met on 52%
median deviation 21.7%

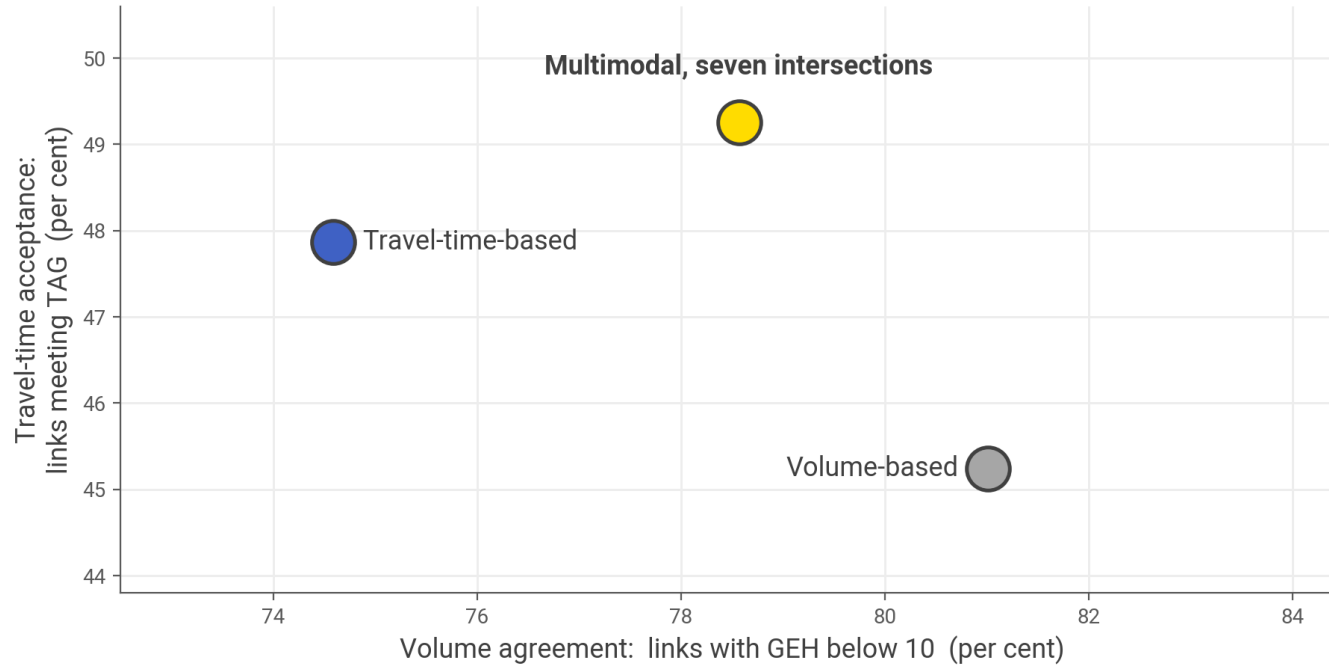


Evening peak

16:00 – 17:00

Criterion met on 48%
median deviation 22.5%

Model comparison



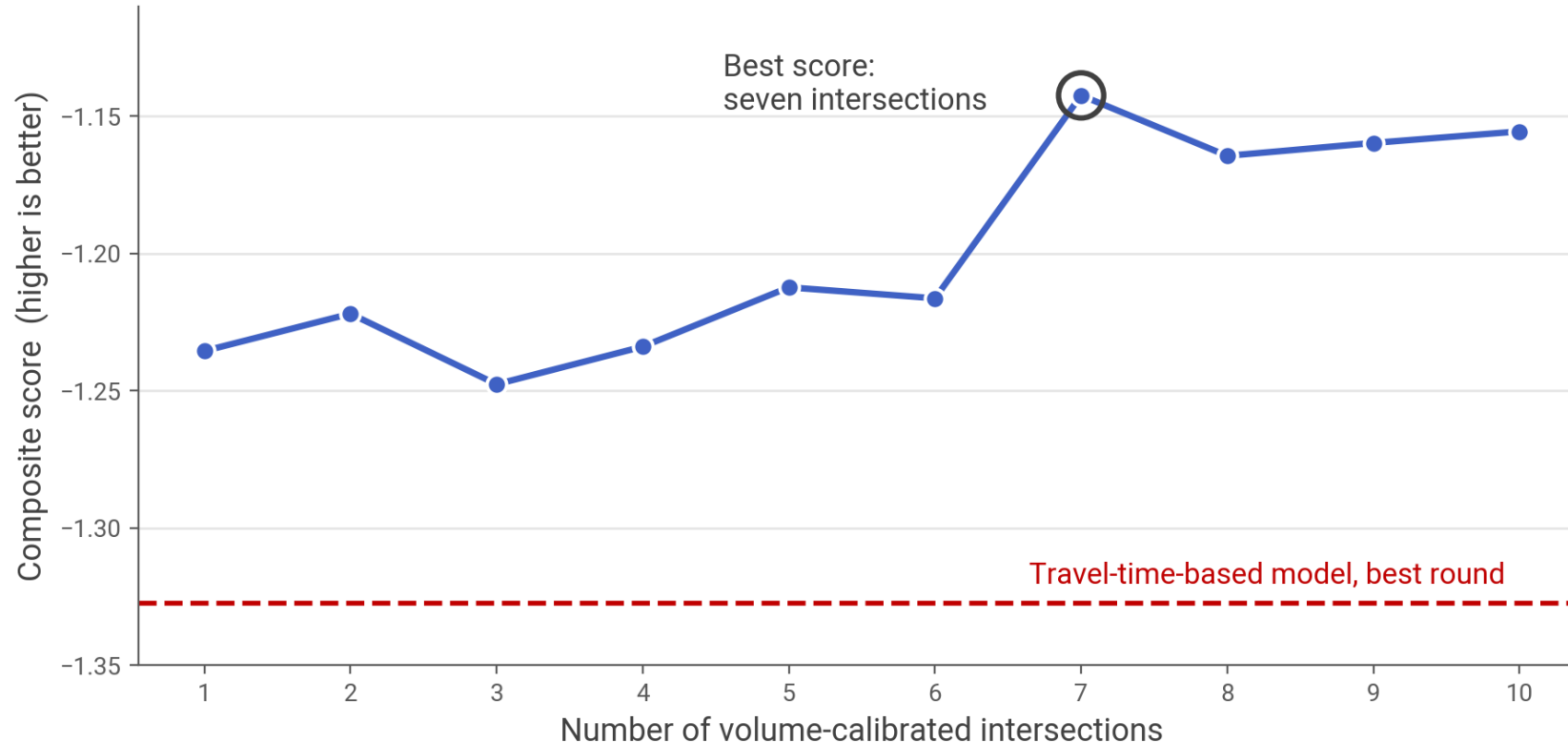
	Travel time	Volume
Volume-based	45.24%	81.01%
Travel-time-based	47.87%	74.59%
Multimodal, seven	49.26%	78.57%

The only model not weakest on either criterion.

Highest travel-time acceptance of the three

Recovers three fifths of the volume agreement the travel-time-based model gives up

Intersection sweep



Below seven, every step pays.

Each intersection added to the volume set lifts the composite score.

One to seven is worth +0.0931.

Beyond seven, the gains stop.

Campaigns of eight, nine and ten all score lower than seven does.

Seven also needs less field data.

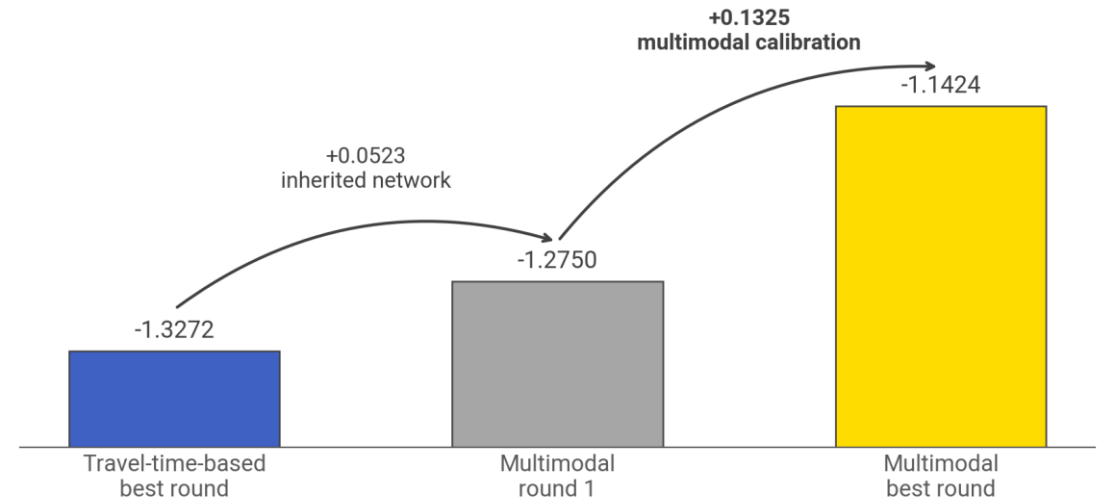
29 volume-calibrated approach links rather than 39, leaving ten more links on the travel-time evidence.

Multimodal results

	Travel-time-based	Multimodal	Change
Composite score	-1.3272	-1.1424	+0.1848
Weakest mode-peak	41.38%	44.83%	+3.45
Average mode-peak	47.87%	49.26%	+1.40
Morning total	51.72%	52.87%	+1.15
Evening total	48.28%	49.43%	+1.15

Better on every measure.

Largest gain on the weakest mode and peak: +3.45.



Where the gain comes from.

28% the inherited network, 72% the multimodal rounds.

Data collection

Video counts

Collected site by site

In Meerut: two thirds of the data budget.

Crowdsourced travel time

Procured, not surveyed

One purchase, no field crews.

Target the spend

Video where queueing dominates, travel time elsewhere.

In Meerut, seven intersections of twenty-eight.

Rank without counting

The ranking comes from the travel-time data.

What it buys

Most of the volume agreement, from a fraction of the sites.

In Meerut, within 2.44 points of counting at all 28.

Thank you!



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